

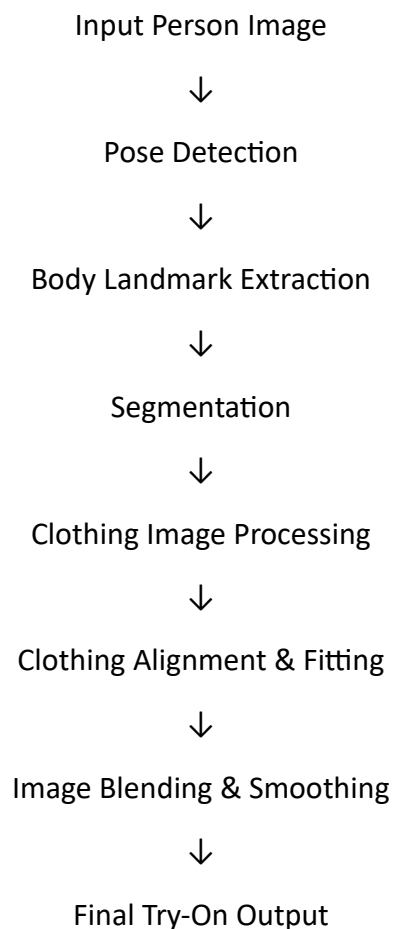
DAY – 12

AI Workflow Documentation

In this task, I documented the workflow of the Virtual Try-On system to understand how different AI components work together to produce a virtual clothing try-on result. Workflow documentation is important because it helps explain the sequence of operations, system architecture, input/output flow, and challenges faced during implementation.

The main objective of this task was to document the complete try-on pipeline, starting from input image collection to pose estimation, segmentation, clothing alignment, and final output generation. This documentation also helps identify limitations in the system and possible improvements for future optimization.

Current Virtual Try-On Workflow



AI Pipeline Structure

The system uses multiple AI and computer vision techniques to simulate clothing placement.

Main components of the pipeline:

1. Person image input
2. Clothing image input
3. Pose estimation and landmark detection
4. Segmentation and masking
5. Clothing resizing and alignment
6. Garment overlay and blending
7. Final visualization output

AI models and tools explored:

- MediaPipe for body landmark detection
- OpenPose for pose estimation experiments
- rembg for segmentation and background removal
- OpenCV for image processing and overlay operations

INPUT/OUTPUT FLOW:

The system workflow follows a structured input-processing-output model to generate a Virtual Try-On result.

The input stage includes a person image, a clothing image, and body pose information used for analysis and fitting.

During the processing stage, landmark detection is performed to identify important body keypoints such as shoulders, torso, and hips for accurate garment placement. Body segmentation and clothing masking are applied to isolate relevant regions and improve fitting quality. The garment is then resized, scaled, and aligned according to body proportions, followed by overlay and smoothing techniques to improve visual appearance and reduce fitting errors.

Finally, the output stage generates a Virtual Try-On image in which the selected garment is virtually fitted onto the person for visualization.

Challenges Faced

During the workflow implementation, a few challenges were observed. Sometimes the clothing did not align properly for different body types and side poses, which affected fitting accuracy. In some cases, background complexity caused poor segmentation, and edge artifacts appeared around garments, making the try-on output look less realistic. Heavy models also increased processing time and reduced overall speed.

Possible Solutions

To improve the system, better pose detection and segmentation techniques were explored for more accurate garment placement and cleaner output. Lightweight AI models were considered to improve processing speed, while dynamic garment scaling based on body landmarks helped improve fitting quality. Smoothing and blending techniques were also used to make garment edges appear more natural and realistic.

Conclusion

Through this task, I documented the complete workflow of the Virtual Try-On system and understood how pose detection, segmentation, garment alignment, and image processing work together to generate try-on results. Workflow documentation also helped identify challenges and possible improvements, making the system easier to understand, optimize, and expand in future development